

MD. SABBIR HOSSEN

📍 Kishoreganj, Dhaka, Bangladesh

☎ +880 1712 xx9 227 ✉ sabbir.hossen@bu.edu.bd

GitHub ✧ Personal Website ✧ Google Scholar ✧ LinkedIn

RESEARCH INTERESTS

Multimodal and Foundation Models

- Large Language Models (LLMs)
- Vision-Language Models (VLMs)
- Visual Representation Learning
- Multimodal Representation Learning

Efficient and Resource-Constrained AI

- Efficient Deep Learning Architectures
- Low-Resource and Multilingual Language Processing
- Model Compression and Optimization

EDUCATION

Bangladesh University

Bachelor of Science

Department of Computer Science and Engineering

Mohammadpur, Dhaka-1207, Bangladesh

January 2020 - December 2023

CGPA: 3.66/4.00

Thesis: A Sophisticated Feature Vectorization-Based Stacked Machine Learning Approach for Fake News Detection in Bangla and English.

EXPERIENCE

Research Assistant

Bangladesh University.

May 2023 - July 2026

Mohammadpur, Dhaka-1207, Bangladesh

- Collected, processed, and analyzed datasets to support machine learning research.
- Designed and implemented machine learning models for research projects in NLP and computer vision.
- Conducted literature reviews and wrote several research papers.
- Served as corresponding author for multiple research papers, handling reviewers' responses and revisions.
- Mentored three undergraduate research groups in research methodology and manuscript preparation.
- Collaborated and coordinated with faculty, researchers, and students from different universities and countries.

PUBLICATIONS

Md. Sabbir Hossen, Fahim Al Farid, Pabon Shaha, Md. Mowahibur Rahman Twake et al. "A Sophisticated Feature Vectorization-Based Stacked Machine Learning Approach for Fake News Detection in Bangla and English." *Social Network Analysis and Mining*, 2025. <https://doi.org/10.1007/s13278-025-01552-6>

Md. Sabbir Hossen, Md. Saiduzzaman, and Pabon Saha. "Social Media Sentiments Analysis on the July Revolution in Bangladesh: A Hybrid Transformer Based Machine Learning Approach." *In Proceedings of the IEEE 17th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, 2025

Pabon Saha, **Md. Sabbir Hossen**, Md. Ibrahim Hosen Sojib, and Sanjida Akter et al. "StackTrace-AI: Identifying Generative AI Text Origins using Ensemble Learning." *In Proceedings of the IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025

Md. Sabbir Hossen and Md. Saiduzzaman. "TransCNN: A Hybrid CNN-Transformer Synergy for Reliable Deepfake Forensics." *In Proceedings of the IEEE 28th International Conference on Computer and Information Technology (ICCIT)*, 2025

Eshat Ahmed Shuvo, Md Shuvon, Nazmul Sarkar, and **Md. Sabbir Hossen** et al. "Optimized Hybrid Cascaded Approach for Accurate Oral Cancer Detection in Histopathology Images Using Deep CNNs." *In Proceedings of the IEEE 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, 2025

Md. Sabbir Hossen, Md. Saiduzzaman, Pabon Shaha, and Mostofa Kamal Nasir. "Jellyfish Species Identification: A CNN-Based Artificial Neural Network Approach." *In Proceedings of the IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2025

Md. Sabbir Hossen, Pabon Saha, and Md. Saiduzzaman. "A Hybrid Machine Learning Approach Utilizing CNN Feature Extraction with Traditional Classifiers to Identify Strawberry Leaf Diseases." *In Proceedings of the IEEE 4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025

Md. Emon Akter Sourov, **Md. Sabbir Hossen**, and Pabon Shaha et al. "Explainable Machine Learning Framework for Cardiovascular Disease Diagnosis and Prognosis." *In Proceedings of the IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE)*, 2026

Md. Sabbir Hossen, Pabon Shaha, and Md. Saiduzzaman et al. “An Explainable AI Driven Machine Learning Approach for Maternal Health Risk Analysis.” *In Proceedings of the IEEE 27th International Conference on Computer and Information Technology (ICCIT)*, 2024

SUBMITTED MANUSCRIPTS

Md. Tanvir Ahmed Shaikat, Pabon Shaha, **Md. Sabbir Hossen**, S M Rifatur Rana et al. “Automated Scoring of Bangla Student Responses: An Efficient and Explainable Deep Learning Approach,” 2026 *[Submitted to a peer-reviewed Journal]*

Pabon Saha, **Md. Sabbir Hossen**, Mostofa Kamal Nasir, Anichur Rahman et al. “Catching the Bots: A Transformer-Based Ensemble Learning for Machine-Generated Text Detection,” 2026 *[Under Review in a peer-reviewed Journal]*

Md. Sabbir Hossen, Eshat Ahmed Shuvo, Shibbir Ahmed Arif, Pabon Shaha, and Anichur Rahman et al. “An Efficient Deep Learning Framework for Brain Stroke Diagnosis Using Computed Tomography (CT) Images,” 2025 *[Under Review in a peer-reviewed Journal]*

Eshat Ahmad Shuvo, Wahidur Rahman, Pabon Shaha, and **Md. Sabbir Hossen** et al. “Optimized Hybrid Approach for Early Detection of Alzheimer’s Disease Using Machine Learning and Deep Learning Techniques,” 2025 *[Under Review in a peer-reviewed Journal]*

Mohammad Shohel Parves, **Md. Sabbir Hossen**, Pabon Saha, and Bikash Kumar Paul et al. “ConvNet9: A Cutting-Edge Customized Convolutional Neural Network Model to Identify Potato Leaf Disease with Web Application,” 2025 *[Under Review in a peer-reviewed Journal]*

CONFERENCE PRESENTATIONS

Oral Presentation: 28th IEEE ICCIT, Cox’s Bazar, Bangladesh, 2025

Oral Presentation: 17th IEEE ECAI, Bucharest, Romania, 2025

Oral Presentation: 4th IEEE ECCE, Chittagong, Bangladesh, 2025

Oral Presentation: 2nd IEEE QPAIN, Rangpur, Bangladesh, 2025

Oral Presentation: 27th IEEE ICCIT, Cox’s Bazar, Bangladesh, 2024

ACADEMIC SERVICE

Journal Reviewer: Journal of Plant Diseases and Protection (JPDP), 2026

Conference Reviewer: IEEE USBEREIT, 2026

AWARDS & ACHIEVEMENTS

Best Technical Presentation, 27th International Conference on Computer and Information Technology, 2024

1st Runner Up in Project Showcasing, Robo Carnival, BUET, 2023

PROFESSIONAL SKILLS

Programming Languages: Python, C, C++

Python Libraries: NumPy, Pandas, scikit-learn, Matplotlib

ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers

Software Tools: Git, GitHub, Microsoft Office, LaTeX, Mendeley

EXTRA-CURRICULAR ACTIVITIES

iTech Expo associated with Techfest IIT Bombay, IUBAT	<i>November 2022</i>
Presented a project on Remote Control Fire Fighter Robot.	
Robo Carnival, BUET	<i>January 2023</i>
Presented a project on Green Energy E-Bike.	
BEC Robo Mania, BUBT	<i>January 2023</i>
Presented a project on an Integrated Smart Home.	

TEST SCORE

IELTS **7.0**